

MOHAMED ABDALLA AWYES ABDALLA

System Engineer – Telecommunications & Railway Signalling Systems

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 Mohamed Aweyes Website

CORE SKILLS

Telecom Systems Engineering • ELV & Low Current Systems • SCADA & RTU Systems • PLC / HMI Programming • IP Telephony & Dispatching • RF & DMR Systems • Network Design & Configuration • Network HCIA/CCNA • CBTC Railway Signaling Systems • BIM • Fiber Optic Technology • AutoCAD • Revit • PCB Design • Embedded Systems • Programming: C, C++, C#, Python • Microsoft Office

PROFESSIONAL SUMMARY

Technical Office & Systems Engineer with more than 5 years of experience in design, technical office engineering, installation, testing, and commissioning of ELV, ICT, Telecom, SCADA, and Railway Signaling support systems for infrastructure, metro, and industrial projects. Strong expertise in preparing design specifications, BOQ, calculation notes, book wiring, shop drawings, and coordination using Revit and BIM, as well as FAT, SAT, ITP, and RAMS documentation. Proven ability to coordinate with offshore teams and multidisciplinary stakeholders to deliver complex projects.

PROFESSIONAL EXPERIENCE

Technical Office Engineer – Hitachi Rail

03/2026 – Present

- BIM Coordinator responsible for hosting telecom and signalling elements for multidisciplinary coordination, implementing first-fix requirements, creating and managing BIM issues, and reviewing models through ACC.
- Prepared comprehensive project documentation & layout, including HLD, LLD, RAMS, FAT, SAT, calculation notes, PDD Layout, Shopdrawing (FDD) layout, method of installation, book wiring, and BOQ (LOM) documentation.
- Coordinated with the offshore teams in Italy and Canada.

Alexandria – Abu Qir metro CBTC Project

- Design All ELV/ICT systems: Fire Alarm, CCTV, PAS, SCADA, PID, Access Control, Broadband System Structured Cabling.
- Prepare design specs, BOQ, calculation notes, book wiring for ELV systems.
- BIM Coordinator responsible for BIM/Revit coordination and clash detection with Civil/Architectural disciplines and all other project entities, including openings, shaft requirements, room sizing, and cable tray routing. Experienced in multidisciplinary coordination and effective clash resolution.
- Coordinate with signaling team for book wiring (Zone Controllers, Object Controllers, Field Element) for CBTC interfaces.
- Designed the Shopdrawing of the CBTC signalling system and the telecommunications system.

Cairo Metro Line 1 – Brownfield CBTC Modernization

- Full ELV survey at 35 stations and depot; participate in design specification phase.
- upgrade/modification strategies for existing infrastructure using BIM coordination and .
- Design All ELV/ICT systems: Fire Alarm, CCTV, PAS, SCADA, PID, Access Control, Broadband System Structured Cabling.
- Prepare design specs, BOQ, calculation notes, book wiring for ELV systems.
- BIM Coordinator responsible for BIM/Revit coordination and clash detection with Civil/Architectural disciplines and all other project entities, including openings, shaft requirements, room sizing, and cable tray routing. Experienced in multidisciplinary coordination and effective clash resolution.
- Designed book wiring diagrams for the CBTC signalling system and the telecommunications system.

System Engineer - ACTEL Communication Egypt

12/2022 – 02/2025

- Executed projects across full lifecycle: design, installation, testing, commissioning, and handover.
- Prepared comprehensive project documentation, including HLD, LLD, RAMS, FAT, SAT, calculation notes, configuration documents, method of installation, book wiring, and BOQ (LOM) documentation.
- Delivered telecom and control systems for railway and infrastructure projects.

Naga Hamady – Luxor Corridor (Railway Project)

- Designed and implemented IP Telephony using Avaya G450/G430.
- Installed and configured Eventide Nexlog 740DX recording system.
- Implemented and configured the Bodet master clock system, including both analog and digital clocks.
- Designed Zetron DCS 5020 dispatching system.
- Designed the architecture and topology of the L2, L3 network system.
- Configured Huawei switches and routers across the entire corridor.
- Designed IP plan Document.
- Implemented signaling communication using Huawei MUX 6608/6616 via microwave and fiber.
- Performed Site Acceptance Tests (SAT) with the consultant and client for the CTC/BTC and two EIS stations.

SCADA System – ENR

- Designed SCADA architecture using DPS RTU TMON Mini.
- Integrated temperature, humidity, smoke, door contact, IR presence, water leakage, and AC monitoring sensors.
- Implemented NMS client-server for RTU monitoring.
- Configured VHF alarms (Forward Power Low, High VSWR).

RF Engineer – ACTEL Communication Egypt

05/2022 – 12/2022

Cairo Metro Line 1 & 2

- Designed and installed DMR mobile radios and base stations.
- Configured dispatching systems.
- Performed RF surveys and coverage testing.
- Configured call types: Group, Broadcast, Private, PSTN, DGNA.

Telecom Site Engineer – ACTEL Communication Egypt

12/2021 – 05/2022

Adly Mansour Transportation Hub

- Installed/commissioned Fire Alarm, VESDA, CCTV, Access Control, PAS, Fiber.
- EASE coverage studies; Cisco switches & IP networks (CCNA level); vendor coordination.

Embedded Systems Engineer – Egy Solar System

05/2020 – 12/2020

- Developed solar tracker prototype; built drone prototype during military service.

TECHNICAL TOOLS

PLC Platforms: Siemens S7-300/400/1200/1500
HMI/SCADA: WinCC V7.2, TIA Portal, DOPLSOFT, VIJEO, NEXION
PLC Software: WPLSoft, ISPSOFT, TIA Portal
BIM/Design: Revit, AutoCAD, Navisworks (clash detection)

CERTIFICATIONS

- PLC Programming Level II (Siemens)
 - HCIA – Huawei Certified ICT Associate
 - Introduction to IoT
 - SCADA & Machine Control
- CCNA – Cisco Certified Network Associate
 - Embedded Systems Diploma
 - Low Current Systems Diploma

INTERNSHIPS

- Petrogulf Masr
 - East Zait Petroleum
 - Banha Factory
- Gemsa Petroleum
 - Rashid Petroleum Company

EDUCATION

Bachelor of Electrical Communication & Electronics Engineering
Egyptian Academy of Engineering & Advanced Technology (2015–2020)
Graduation Project: Grass-Cutting Automation Robot (Excellent)

LANGUAGES & SOFT SKILLS

- **Arabic** – Native
 - **English** – Very Good
- Soft Skills:** Problem Solving, Project Management, Team Leadership, Technical Documentation, Communication, Time Management